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RESEARCH ARTICLE

# Farmers' preferences for the next generation of maize hybrids: application of product concept testing in Kenya and Uganda

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**Abstract**
Step-change innovation in seed product design by public sector crop breeding has led to major contributions to global food security. The literature, however, provides few insights on how to identify forward-looking innovation opportunities. Inspired by discussions in the product innovation literature, this article describes our application of product concept testing in the context of hybrid maize in Uganda and Kenya. We identified the following eight maize seed product concepts based on interactions with seed companies, crop breeders, and farmers: 'Resilience', 'Drought escape', 'Food and fodder', 'Home use', 'Green maize', 'Livestock feed', 'Intercropping', and 'Family nutrition'. These were described and presented to 2400 farmers using videos, where each farmer saw three concept-presentation videos. Farmers were most likely to have selected the resilience (Kenya and Uganda), drought escape (Uganda), and intercropping (Kenya) concepts. Farmers showed mixed interest in other concepts, such as home use and food and fodder, suggesting that investments in product production and promotion would be required in addition to investments in breeding. These results provide new entry points for conversations among transdisciplinary teams at regional and national levels on the current and future opportunities for crop breeding to respond to farmers' requirements for new seed products.

**Keywords:** Hybrid maize; crop breeding; food security; market intelligence; intercropping; tricot approach

## Introduction

In response to the challenges posed by climate change, population growth, and rural poverty, crop breeding programs in the global South must deliver new seed products that meet the requirements of famers, processors, and consumers. In general, this implies technologies that resist pests and diseases with fewer inputs and increase production and productivity by maximizing resource use efficiency of water, land, and nutrients (Jorasch 2019; Evans & Lawson 2020). Crop breeding programmes may also need to incorporate a host of other requirements into product design related to how crops are grown and used. Farmers may require products, for example, that respond to the needs of processors and traders (e.g. crop parameters related to colour, size and taste), result in less drudgery for home processing and cooking, or complement different aspects of on-farm production systems (e.g. irrigation, dual purpose, and intercropping). Moreover, farmers' requirements for seed evolve over time and vary in nature, in response to contextual

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differences, to include production systems, gender relations, socioeconomic conditions, and market opportunities.

Researchers have highlighted priorities and advances in crop breeding for yield stability under environmental and biological stresses (Cairns *et al.* 2013), as well as the potential for new technologies to accelerate innovation in seed-product design, such as high-throughput sequencing (Sinha *et al.* 2021; Cvejic *et al.* 2022), gene editing (Flavell 2017), and genetic modification to accelerate innovation in seed product design. The potential for breeding to deliver crops with enhanced nutritional benefits has been discussed at length (Saltzman *et al.* 2017; Maqbool & Beshir, 2019; Sinha *et al.* 2021). Recently, a consultation among nearly 600 experts yielded several trait priorities for six crops in addition to yield gain, from optimized rhizosphere microbiome to improved stover quality and increased early vigor (Pixley *et al.* 2023). Given the gradual decline in funding for agricultural research, a fundamental challenge lies in directing scarce resources towards new seed products with the greatest chance of delivering impact. In this context, a critical question is what the next-generation seed products are that farmers and others (e.g. final consumers and food processors) require and that farmers are willing to seek out and grow in their fields.

The history of crop breeding for farmers in the global South contains multiple examples of radical, step-change innovation in seed product design. The innovation responded to a thorough understanding of farmers' requirements in a specific production context (Byerlee & Dubin 2010; Lumpkin 2015) and the potential positive impact on final consumers and the environment. Notable examples include semi-dwarf wheat and rice, orange-fleshed sweet potato, Bt cotton, and hybrid sorghum. These achievements rested on innovations in breeding technology, as well as the vision, energy, connections, persistence, and perhaps luck of a few individuals and their teams. As explained by Low and Thiele (2020, p. 1), 'the development and scaling of orange-fleshed sweet potato (OFSP) during the past 25 years is a case study of a disruptive innovation to address a pressing need – the high levels of vitamin A deficiency among children under five years of age in sub-Saharan Africa'. A common theme among the major crop breeding successes is the step-change innovation inherent in seed product design that was apparent and relevant to farmers and other market actors (e.g. traders, processors, and final consumers).

However, the definition of success in publicly funded crop breeding, to include breeding targets, has tended to focus on incremental yield gains and increased pest and disease resistance under rainfed conditions (Masuka *et al.* 2017; Tadesse *et al.* 2019; Adhikari *et al.* 2021; Batte *et al.* 2021; Asea *et al.* 2023). These targets reflect the influence of major donors (Tarjem *et al.* 2022) looking to address food security concerns, as well as the internal interests and histories of crop breeding programs. Social science research has tended to validate the prioritization of breeding efforts around yield gains and yield stability (Ajambo *et al.* 2017; Kassie *et al.* 2017; Marenya *et al.* 2022). In the case of maize farmers in East Africa, the majority apply few external inputs and face large yield gaps (Leitner *et al.* 2020), and thus are unlikely to benefit from the incremental gains in productivity derived from maize breeding. Farming households in Kenya and Uganda tend to pursue diversified livelihood strategies that incorporate agriculture production (e.g. maize and legumes), livestock production, and off-farm employment (Valbuena *et al.* 2015). Maize plays an important role in home consumption, as well as in income generation. From a seed product design perspective, this implies farmers have requirements related to consumption (e.g. taste and dry milling quality) and production (e.g. yield stability) (Almekinders *et al.* 2021).

Various studies have highlighted the disappointingly slow uptake of improved varieties across crops and geographies (McEwan et al 2021; Walker & Alwang 2015). In the case of maize in East Africa, impact from maize breeding programs has been thwarted by slow turnover, i.e. farmers' tendency to purchase older products over newer ones, despite the potential advantage in yield that the newer products offer (Rutsaert & Donovan 2020). Recently, the incorporation of concepts, such as 'demand-led breeding', into the public-sector crop breeding lexicon (Anthony 2013; Persley & Anthony 2017), has signaled the objective to achieve more impact by addressing the

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supply-side bias in how breeding pipelines have been designed and prioritized – i.e. where investments among crops and within pipelines tend to reflect donor and breeder assumptions about farmer and end-user requirements (Tarjem *et al.* 2022). In their review of literature, Thiele and colleagues (2020, p. 1076) noted 'a strong confirmation for the hypothesis that insufficient priority given to consumer-preferred traits by breeding programs contributes to the limited uptake of modern varieties and low varietal turnover'. Recent studies have highlighted the potential for breeding programs to respond to consumer demands (Arnaud et al. 2024; Vu *et al.* 2023). As a concept for guiding discussions on future seed product design, however, demand-led breeding lacks clarity on whose demand matters for crop breeding – breeding programs could prioritize someone's demand (e.g. final consumer or processor), resulting in products that fail to meet the expectations of the farmers who take the decisions on which seeds and other inputs to acquire.

Looking back, social science discussions on farmers' requirements for improved varieties have tended to take place within the guardrails of established breeding priorities. In the 1980s, for example, discussions emerged on participatory plant breeding (PPB) – an approach designed to shift crop breeding programs towards more local level by directly involving farmers in the product-design process (e.g. Ceccarelli & Grando 2007; Weltzien & Christinck 2017). As described by Ceccarelli and Grando (2007, p. 145), PPB is a process whereby 'genetic variability is generated by breeders; selection is conducted jointly by breeders, farmers, and extension specialists in a number of target environments; and the best selections are used in further cycles of recombination and selection'. Proponents argued that PPB made varietal development efforts more responsive to the needs of technology users, particularly members of rural households that were not well integrated into the market economy. However, a key element that determines the innovation potential is when, how, and under what conditions farmer participation in breeding occurs. McGuire (2008, p. 139) noted that participatory plant breeding restricted farmers' inputs to product design: 'early choices around agroecological classifications, germplasm use, and F1 hybrid development became "locked-in", consequently resisting change. Technical constraints, breeding routines, and actor networks all reinforce particular choices from the past . . . '

This article takes inspiration from discussions in the product innovation literature on how to understand consumer requirements, engineer products that respond to those requirements, and position the resulting new products in the market (Griffin & Hauser 1993). With a bit of imagination, one can see the similarities between 'consumer' and 'farmer', 'engineer' and 'breeding program', and 'market' and 'seed system'. Iuso (1975) introduced the ideas and methods behind concept testing to understand and prioritize customer requirements for new products. Consumers are presented one or different product idea(s), or concept(s), with the goal to obtain their feedback as well as their interest to purchase and use the product if it were to become available. Concepts may be presented in the form of a text description, picture, video, or prototype. The testing takes place early in the product innovation cycle before scarce R&D resources are committed. The insights gained are used to guide discussions on refinements in product design and future investments in production and marketing. While product concept testing has been applied for long in the commercial sector and food sector (e.g. Leek, Maddock & Foxall 1998; Jones & Jew 2007; Gil-Pérez *et al.*, 2023), it has seen limited application by public sector research programs.

Government, donor, and private sector investments in maize breeding across much of the global South are insufficient to design, test, and produce the range of products that would meet all the current and future requirements of farmers and end users. Therefore, this article aims to provoke new discussions among breeders, donors, and seed companies on options to prioritize investments for future products. We employed concept testing to identify promising new maize hybrids based on different farmer and end user requirements in the western Kenya and central Uganda. Given the obvious challenge of producing prototypes of new seed products, we employed a novel approach to concept presentation. Our application, which we refer to as video-based product concept testing (VPCT), used videos to describe the uses of different potential types of

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maize hybrids. The remainder of this article is organized as follows. Section 2 presents the methods we employed to identify product concepts, translate the concepts into videos, and design interview protocols and sample frame. Section 3 presents the sex-disaggregated results in terms of product concepts most likely to be preferred by farmers. We conclude in section 4 with implications of the results for maize breeding and efforts by social science to engage on discussions on step changes in seed product design.

## Product concepts and sampling strategy

### *The setting*

The maize seed landscape in Kenya and Uganda contains, respectively, 70–80 and 20–30 products, the majority of which are white hybrids designed for rainfall production environments. Dozens of local, as well as multinational seed companies, produce hybrid maize seed (Mastenbroek, Sirutyte & Sparrow 2021; Spielman & Smale 2017). A network comprised of hundreds of mainly small-scale, independently owned agrodealers (Barriga & Fiala 2020; Mabaya *et al.* 2021; Rutsaert & Donovan, 2020) sell maize seed products to farmers across the two countries. A major distinguishing factor in terms of seed product design is the maturity level (Rutsaert *et al.* 2023). CGIAR and NARES maize breeding programs target the following maturity levels for mid-altitude production regions: early, intermediate, and late maturity products for Kenya and intermediate and late for Uganda. During the long-rains season farmers in high altitudes are likely to benefit most in terms of yield from late-maturity maize, while those in the mid (low) altitudes are likely to benefit most from intermediate (early)-maturity products.

### *Study design*

The first step was to ideate the concepts of maize seed products for farmers who produced hybrid maize, either for the market and for their own consumption, under rainfed conditions, and who were in production zones classified as wet intermediate (receiving over 600 mm rainfall in a season). We identified eight maize seed product concepts and translated these concepts into a video format for presentation to farmers. We presented these concepts to 2400 maize farmers in western Kenya and central Uganda during a short household survey. The following subsections detail the how we (i) designed the concepts, (ii) translated the concepts into a video format, and iii) engaged with farmers for concept evaluation.

#### *Product concepts*

Product concepts were ideated based on insights gained from key informant interviews with representatives from seed producing companies (n = 4) and retailers (n = 6), as well as plant breeding institutes (n = 5), and semi-structured focus group discussions in Kenya (n = 6) and Uganda (n = 6). Focus group discussions involved 6-9 farmers, with separate groups for men and women. Key informant interview lasted on average 1 hour and the focus group discussions 1.5 hours. Table 1 summarizes the discussed topics with each of those stakeholder types. We sought to capture the breath and diversity of maize seed requirements across the different stakeholder types. Farmer and agrodealer discussions focused on the (changing) role of maize in farmers' livelihoods, how maize was consumed and taste, size, and colour expectations, growing practices, grain storage, and yield expectations. Discussions with private seed company experts and plant breeders covered the technological possibilities of breeding, trends in farmer purchases of maize seed products, and the suitability of other types of maize that were not frequently grown in East Africa, such as single-cross white maize hybrids and yellow maize for the livestock industry.

Table 2 presents the eight seed product concepts that emerged from our discussions with maize industry stakeholders. The description of each concept included an explanation of how the

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**Table 1.** Discussion points with key informants for product concept ideation

| | Key informant type | | | |
|---|---|---|---|---|
| Discussion point | Farmers | Agro-dealers | Seed companies | Plant Breeders |
| Changing role of maize in farmer livelihoods | X | X | | X |
| Growing practices | X | X | X | X |
| Grain use in the household and sales | X | X | | |
| Strengths and weaknesses of the current maize seed offer | X | X | X | |
| Technological advancements in maize breeding | | | X | X |
| New market trends | | X | X | X |
| Evolutions in maize markets in other regions | | | | X |

**Table 2.** Product concepts for hybrid maize seed in Kenya and Uganda

| Product concept | Main grower or end-use requirement |
|---|---|
| 1. Resilience | Resistance to major stresses including drought, diseases, and low soil fertility |
| 2. Drought escape | Suitable for early harvesting, with reduced risk in production |
| 3. Food and fodder | Optimized for both household food and fodder for cattle |
| 4. Home use | Optimized for home-use traits such as taste, flour conversion, and storability |
| 5. Green maize | Needs of the fresh green-maize market met (including sweet taste and large cobs) |
| 6. Livestock feed | Reliable feed for chicken, pigs, and other animals |
| 7. Intercropping (with beans) | Optimal yield when maize rows are alternated with beans |
| 8. Family nutrition | Provitamin A – rich variety that provides a key nutrient for young children |

product responded to potential requirements for maize production, maize consumption, or the farm-level production system. For each concept, we provided relevant information about several standardized parameters including yield potential, fertilizer needs, colour, maturity period, and grain usage. There was also general information given in an introduction video that applied to all concepts such as insurance that all the products described in the concepts had good germination rates and were resistant against most common diseases The full descriptions are available in the Supplementary Material Table S1.

*Resilience*

The concept focuses on yield stability in the face of heat and water stress. During seasons where rainfall and heat levels follow historical trends, expected yields would be at par with other hybrid seed products. However, during years where rainfall and heat fall below (above) historical trends, the yields obtained from this product concept would likely be higher than from others hybrid seed products.

*Drought escape*

Unreliable rainfall patterns are a major concern for maize production without irrigation in East Africa (Cairns *et al.* 2013; Wainwright *et al.* 2019; Kogo *et al.* 2022). Drought escape focuses on the early harvest of maize, which reduces the risks of drought during the late stage of production (Chaves *et al.* 2003). Farmers might also seek out early maturity products to i) fetch higher market prices by selling early when prices are highest and ii) attempt to liberate planning space for another crop. While public breeding programs invest in the design and distribution of early maturity products, the focus has been on farmers in dryland areas. Whether farmers require early maturity in areas where rainfall and heat patterns have traditionally allowed for longer maturity periods remains an open question.

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*Food and fodder*

A dual-purpose concept may appeal to farmers who aim to prioritize both grain *and* fodder (particularly stover), and typically the full (above ground) biomass. The concept suggests that breeding investments seek to optimize high fibre in the leaves and stems besides traditional investments in high and stable yields, which reduces the focus on yield. However, the widespread use of maize stover suggests it indeed has value, although typically less than the maize grain (Blümmel, Grings & Erenstein 2013). This concept speaks to the prevalence of mixed crop–livestock systems in Kenya and Uganda (Thornton & Herrero 2015).

*Home use*

This concept responds to farmers who prioritize maize gain for household consumption that contains preferred end-use attributes. Maize production for home use is a requirement that is often stated to be more of interest to women (Weltzien *et al.* 2020). The popular seed product H614D, released in 1986 by the Kenyan Seed Company, provided inspiration for this concept: Christinck *et al.* (2018) explained that its popularity can be partially explained by its taste as well as its hard grain which make it optimal for storage given its resistance to mold and insect damage.

*Green maize*

The green maize concept recognizes the popular consumption habit in Kenya and Uganda of eating roasted maize (Ekpa *et al.* 2018). Basic requirements for roasted maize are large cobs and sweet flavor. Roasted maize offers many of the nutritional benefits of raw maize, to include crude protein, crude fiber, ash, and carbohydrate content (Ayatse, Eka & Ifon 1983). Differences in kernel characteristics caused by genetic inheritance and harvesting time can influence the processing, utilization, and consumption experience (Oladeji *et al.* 2015). Economically, selling roasted maize on the roadside or in markets has much higher value than selling maize in bulk to traders.

*Livestock feed*

Maize grain is often provided as feed to chicken and pigs. Farmers preferred to use maize as feed rather than sell at low prices in local markets. In Kenya and Uganda, most rural households maintain chickens and the commercial poultry sector has been growing rapidly. In Kenya, for example, the 2014 contribution of poultry offtake and egg production to the national agricultural gross domestic product was estimated at 1.3% (USD 46.2 million) and 2.9% (USD 103.1 million), respectively (KNBS 2015). Our product concept for feed was ideated to contain relatively high levels of protein and carotenoids, which would provide for a yellow coloured yolk (Díaz-Gómez *et al.* 2017). While higher-protein maize products for animal feed have been discussed for many years (e.g. López-Pereira 1993), their application in Kenya and Uganda as product concepts allowed for a forward-looking option for farmers with requirements aimed at responding to the poultry industry.

*Intercropping*

Crop breeding programs typically evaluate crosses under monocropping farming conditions (Isaacs *et al.* 2016). However, farmers often grow maize intercropped with crops such as common bean, pigeon pea, cowpea groundnut, and soybean among others (Isaacs *et al.* 2016; Rusinamhodzi *et al.* 2012; Snapp & Silim 2002). An assessment of LSMS[1] data for Tanzania showed that 81% of all households intercropped at least one plot which maize, with maize being the most frequently used crop to intercrop. Recognizing that the maize–beans combination is the most popular intercropping system in the area, we designed a concept specifically targeted towards the requirements of maize–bean intercropping, where maize would be designed to be less

[1]Living Standards Measurement Study

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competitive, benefit from wider spacing, and provide higher-than normal lodging resistance (Davis & Garcia 1983).

### Family nutrition

Crop breeding for increased pro-vitamin A content in different crops has been a longstanding priority (Ortiz-Monasterio *et al.* 2007) due to micronutrient deficiencies in developing countries especially among low-income groups (West & Darnton-Hill 2008). In maize, fortification with provitamin A carotenoids results in grain that has an orange colour. Studies have shown that consumer acceptance may not be an insurmountable barrier for the spread of orange maize production and consumption (De Groote, Kimenju & Morawetz 2011; Smale *et al.* 2015) – e.g. the consumption of orange maize is popular in rural parts of Southern Africa (Manjeru 2017). Pro-vitamin A maize seed products are not yet available in Uganda or Kenya.

### Translation of concepts into videos

The product concepts were translated into a video format. The use of videos offered three advantages for data collection: (i) eliminated concerns regarding farmer illiteracy, (ii) required a low cognitive effort of participants, thus potentially improving data quality, and (iii) eliminated interviewer bias when presenting product concepts to farmers. The Ugandan videos were made in November 2021 and formed the basis for the Kenyan videos developed in February 2022. The videos for Uganda and Kenya were produced professionally, with realistic agrodealer sets and professional actors.

For each country, 18 videos were produced in the local languages: Luganda for Uganda and Swahili for Kenya. Two videos were produced for each product concept: one with a male actor and another with a female playing the role of agrodealer introducing a new (hypothetical) seed product (Supplementary Material Fig. S1). In addition to the videos that presented product concepts, we produced (i) an introductory video where the actor described the exercise and what was expected of the participant, i.e. that he/she would see three product descriptions in the next videos and (ii) a closing video, where the actor (agrodealer) thanked the participant for their time and described the next steps of the survey, including the ranking exercise.

We designed a similar narrative arc for the presentation of each product concept: an agrodealer sought to inform a farmer about a newly stocked seed product. The actor playing the agrodealer – dressed in a white lab coat who stood behind a counter and in front of various bags of maize seed – presented the product and described its potential use values. Product concept descriptions lasted approximately 90–120 seconds.

### Concept evaluation

Each participant farmer viewed five short videos: the introductory video, three randomly selected product concept videos (out of the eight product concept videos available), and the closing video, for a total of approximately five to seven minutes of viewing time. The use of an incomplete block was inspired by previous studies that applied triadic comparison of technology options (tricot) approach (Van Etten *et al.* 2016). After viewing the videos, we asked the farmers to select their most and least preferred product concepts. After the concepts comparison, the farmers participated in a 30-minute survey that sought information on their socio-demographic profile, farm characteristics, and maize farming, including seed purchase frequency, cropping system, seed quantities and products purchased, and grain harvest volume and use. Data entry was carried out with tablets using ODK (https://getodk.org) powered by the FormShare software (https://formshare.org).

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### *Sampling strategy*

The study was implemented between November 2021 and April 2022 in mid-altitude zones of Kenya (Kakamega and Bungoma counties) and Uganda (Mubende and Mityana districts) (Supplementary Material Fig. S2). A multi-stage stratified sampling technique was employed to select regions, villages, and households, combining purposive and random sampling. Within each district/county three sub-counties were randomly selected. The number of villages (clusters) within each sub-county was determined and randomly sampled with probability proportional to size. The same number of households were sampled from each cluster, irrespective of the cluster size; this resulted in the random selection of 1,200 maize farming households per country and in each household the selection of the male or female respondent was predefined.

### *Data analysis*

The analysis of participants' ranked preferences for product concepts was performed using the Plackett–Luce model (Luce 1959; Plackett 1975), a statistical approach for ranking preferences or choices. This model, based on the Luce's Axiom, estimates the probability (*worth*) of selecting an item (e.g. product concept) over others in a given set. It has been applied in various fields, including sports rankings, comparative judgement exercises (Chambers & Cunninham 2022), consumer preference trials (Olaosebikan *et al.* 2023), and crop variety selection (van Etten *et al.* 2019). Ranking methods in revealed-choice experiments, like consumer preference trials, are rapid, adaptable to local context, and are a relatively easy data collection exercise both for enumerators and respondents, requiring minimal respondent calibration (Coe 2002).

We derived pairwise probabilities from the fitted Plackett–Luce model to assess the relative likelihood of one maize concept being preferred over another. These probabilities represented the estimated chance that participants selected a specific concept when compared directly to other concepts in the dataset. The *worth* parameters estimated by the Plackett–Luce model were transformed into a pairwise comparison matrix, where each cell indicated the probability of preference between two concepts. These probabilities provided a detailed, quantitative comparison of all possible concept pairs, enabling the understanding of relative preferences within concepts.

To account for participant heterogeneity influencing preferences for different concepts, we used Plackett–Luce model combined with model-based recursive partitioning (Zeileis *et al.* 2012). To fit the model, we selected the variables gender, country, and size of farmland as covariates. The recursive partitioning algorithm for the Plackett–Luce model involves four main steps, as described by Turner *et al.* (2020): (i) fit a Plackett–Luce model to the complete dataset; (ii) assess the stability of worth parameters for each covariate; and (iii) if significant instability is detected, split the dataset into two partitions based on the covariate showing the highest instability. The split threshold is chosen to provide the greatest improvement to the model fit. Steps 1–3 are repeated until no further instabilities are detected, or dataset partitions fall below a pre-specified size threshold. This results in a combination of Plackett–Luce models, called 'Plackett–Luce trees'.

Analysis of ranking data with the Plackett–Luce model was performed in R using the packages PlackettLuce (Turner *et al.* 2020), gosset (de Sousa *et al.* 2023), and multcompView (Graves *et al.* 2019).

### *Analysis of open responses*

We performed a text mining approach to analyse participant responses regarding preferences for maize concepts. We first organized and processed the open-ended textual data using R. The dataset was pre-processed by breaking the text into smaller unit responses (tokening), removing stop words, and standardizing text to remove typos. Stop words are common words like 'the', 'and', or 'is' that frequently appear in text but carry little significant meaning in terms of content or

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topic relevance. Removing these words helps reduce noise in the analysis and allows algorithms to focus on more meaningful terms. The R packages tidytext (Silge and Robinson 2016) and tm (Feinerer & Hornik 2020) facilitated text cleaning and tokenization. This step generated cleaned text datasets suitable for downstream analysis.

To explore relationships between words in participants' responses, a correspondence analysis was conducted (Hirschfeld 1935). Correspondence analysis is a multivariate statistical technique used to explore associations between categorical variables, in this case, response patterns and maize concept attributes. The method involves constructing a contingency table of words and concepts and visualizing these relationships in a two-dimensional space. The resulting network provides insights into the cognitive structures and priorities driving participant preferences. The R packages FactoMineR (Le et al 2008) and factorextra (Kassambara and Mundt 2020) were used to perform the correspondence analysis.

All the data and code used for the ranking analysis and text mining are available on GitHub https://github.com/AgrDataSci/maize-concept-testing/.

## Results

### *Sample description*

Table 3 summarizes the characteristics of the sampled farmers. The average age in Kenya and Uganda was, respectively, 49 and 42 years, with women tending to be older in Kenya but younger in Uganda. Most respondents completed at least primary school and 37% in Kenya and 27% in Uganda completed secondary or post-secondary school. Table 4 covers maize production practices of the sample in Kenya and Uganda. Participants in Uganda reported larger farms sizes, 7.4 acres on average, compared to farm sizes in Kenya of 1.7 acres on average. More than half of the Kenyan farm size was used for growing maize with 1.1 acres on average, while this was only 2.5 acres for Uganda. Also in production practices, there were large differences between both countries. While over 90% of male as well as female farmers in Kenya bought seed at least once a year, this was under 50% for farmers in Uganda. Intercropping, mainly with maize and beans, was popular among Kenyan farmers and the reported growing practice by more than 80% of the farmers. In Uganda, monocropping was more prevalent and significantly higher with male compared to female farmers. In both countries, most farmers grow only one seed product. Use of the maize harvest differed between Kenya and Uganda: 88% of the maize harvest was sold in Uganda, while in Kenya the harvest was mainly used for consumption and sales.

### *Product concept evaluation*

We utilize various figures and tables to summarize the results of the concept evaluations. Figure 1 presents the Plackett–Luce estimates (log-*worth*) of the eight product concepts across the entire sample, while Figure 2 conveys farmers' positive and negative open-ended responses associated with the concepts. Table 5 presents the pairwise probabilities obtained from the Plackett–Luce *worth* estimates. Lastly, Figure 3 explores the differences in preferences accounting for country, gender, and farm size as covariates to split the sample in the Plackett–Luce trees.

#### *The top 3: Resilience, Drought escape, and Intercropping*

Across the farmers in Kenya and Uganda, three concepts had a higher probability of being the most preferred concept compared to the other concepts: *Resilient (benchmark)*, *Drought escape*, and *Intercropping*. One element these three concepts had in common was that the main use value focused on the agronomic performance of the maize crop. Figure 2 specifies the reasons why farmers selected their most and least preferred concept. Especially *Drought escape* was perceived as different and was related to words like 'drought', 'fast', and 'shorter' as positive drivers – keywords

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**Table 3.** Participant characteristics

| | Kenya | | | | Uganda | | | |
|---|---|---|---|---|---|---|---|---|
| Variables | Total | Male | Female | p-value | Total | Male | Female | p-value |
| Sample size | 1200 | 607 | 593 | | 1200 | 600 | 600 | |
| Age (Years) | 48.97 | 46.96 | 51.02 | 0.000 | 42.43 | 43.97 | 40.88 | 0.000 |
| | (14.44) | (13.59) | (14.99) | | (13.77) | (14.33) | (13.02) | |
| Education level (%) | | | | 0.000 | | | | 0.004 |
| No formal education | 5.83 | 8.57 | 3.04 | | 13.42 | 10.33 | 16.50 | |
| Primary | 56.41 | 58.98 | 53.80 | | 59.41 | 59.33 | 59.50 | |
| Secondary | 27.50 | 26.52 | 28.50 | | 23.00 | 24.83 | 21.17 | |
| Post-secondary | 10.25 | 5.93 | 14.67 | | 4.17 | 5.50 | 2.83 | |
| Marital status (%) | | | | 0.000 | | | | 0.000 |
| Married | 82.00 | 72.65 | 91.57 | | 81.83 | 88.33 | 75.33 | |
| Not married | 18.00 | 27.35 | 8.43 | | 18.17 | 11.67 | 24.67 | |
| Household size | 6.44 | 6.32 | 6.56 | 0.123 | 6.13 | 6.14 | 6.12 | 0.899 |
| | (2.74) | (2.59) | (2.87) | | (2.73) | (2.85) | (2.60) | |

**Table 4.** Participant maize farming characteristics

| | Kenya | | | | Uganda | | | |
|---|---|---|---|---|---|---|---|---|
| Variables | Total | Male | Female | p-value | Total | Male | Female | p-value |
| Sample size | 1200 | 607 | 593 | | 1200 | 600 | 600 | |
| Total area managed (acres) | 1.69 | 1.52 | 1.86 | 0.000 | 7.43 | 6.19 | 8.66 | 0.462 |
| | (1.55) | (1.43) | (1.65) | | (5.80) | (6.29) | (8.18) | |
| Area under maize (acres) | 1.14 | 1.09 | 1.20 | 0.052 | 2.47 | 2.72 | 2.22 | 0.001 |
| | (1.02) | (1.00) | (1.04) | | (2.44) | (2.83) | (1.96) | |
| Frequency of buying seed (%) | | | | 0.389 | | | | 0.896 |
| Every season | 74.67 | 74.30 | 75.04 | | 18.25 | 19 | 17.50 | |
| Once every year | 22.92 | 22.73 | 23.10 | | 30.75 | 31.67 | 29.83 | |
| Once every two years | 0.58 | 0.66 | 0.51 | | 11.83 | 11.83 | 11.83 | |
| Once every three/four years | 0.5 | 0.33 | 0.68 | | 11.33 | 10.84 | 11.83 | |
| Once every five years or less | 1.33 | 1.98 | 0.67 | | 27.83 | 26.67 | 29 | |
| Growing practices (%) | | | | 0.316 | | | | 0.009 |
| Monocropping | 12.25 | 12.36 | 12.14 | | 62.33 | 66.50 | 58.17 | |
| Intercropping with beans | 70 | 68.2 | 71.84 | | 25.50 | 21.50 | 29.5 | |
| Intercropping with crops/trees other than beans | 1.83 | 2.14 | 1.52 | | 2.33 | 2 | 2.67 | |
| Both monocropping and intercropping | 12.42 | 14.17 | 10.62 | | 9.83 | 10 | 9.67 | |
| Varieties of maize grown (%) | | | | 0.513 | | | | 0.382 |
| One | 70.33 | 70.68 | 69.98 | | 86.25 | 84.67 | 87.83 | |
| Two | 25.50 | 25.04 | 25.97 | | 12.00 | 13.33 | 10.67 | |
| More than two | 4.17 | 4.28 | 4.05 | | 1.74 | 2.01 | 1.50 | |
| Harvest usage (%) | | | | | | | | |
| Sold | 37.16 | 24.62 | 52.51 | | 88.43 | 88.91 | 87.71 | |
| Consumed | 44.16 | 33.91 | 22.40 | | 2.93 | 2.67 | 3.33 | |
| Used as payment | 11.79 | 9.12 | 14.94 | | 0.30 | 0.30 | 0.31 | |
| Given out free | 5.00 | 3.75 | 6.55 | | 0.25 | 0.25 | 0.26 | |
| Animal feed | 1.70 | 1.19 | 2.18 | | 4.90 | 4.63 | 5.33 | |

that were not associated with other concepts. *Intercropping* presented a closer relationship with 'yield', 'soil', and 'fertilizer', and the *Resilient* was also linked to 'yield', 'soil', and 'resistant'.

The pairwise probabilities in Table 5 show that the three concepts had a higher percentage of farmers preferring them over the other five concepts. However, in a direct comparison with each other, there was no clear leader. Approximately half of the farmers preferred *Resilient* over *Drought escape* and vice versa. The same was true for *Resilient – Intercropping* as well as *Drought escape – Intercropping*.

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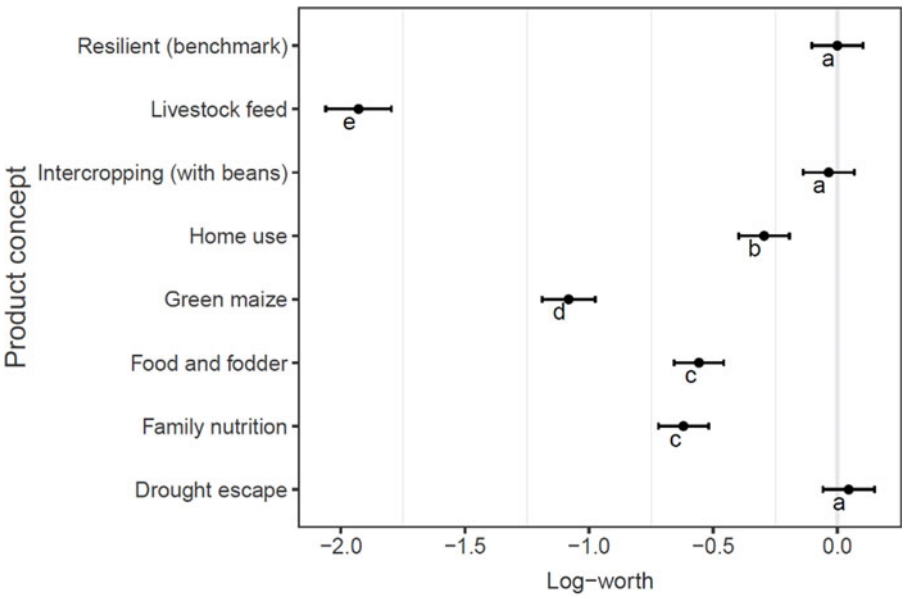



**Figure 1.** Model estimates (log-worth) of farmers' preferences on the product concepts of hybrid maize presented in Kenya and Uganda. *Log worth are coefficients derived from the Plackett–Luce model, which estimates the probability of one given concept in outperforming all the other concepts in the set. The concept 'Resilient (benchmark)' is set as a reference (log-worth arbitrarily set to zero). Different letters indicate significant differences at* $p < 0.05$.

The Plackett–Luce trees, however, revealed that participant heterogeneity (specifically country, gender, and land size), significantly influenced concept selection. Women in Kenya with small plot sizes (< 0.9 acre) preferred the *Intercropping* concept over the *Resilient* and *Drought escape* concepts, while men tended to prefer the *Intercropping* and *Resilient* concepts over the *Drought escape* concept. In Uganda, *Drought escape was clearly preferred over* other concepts for male farmers, while women showed preferences for two products *Drought escape* and *Resilient*.

*In the middle: Home use, Food and fodder and Family nutrition*

*Home use*, *Food and fodder,* and *Family nutrition* followed the three agronomy focused concepts in terms of farmers' preferences. Between the three of them, *Home use* was slightly more preferred.

There was a difference in evaluation of *Home use* between male and female farmers (Figure 3). Women in Kenya preferred the Home use concept equally to *Drought escape* and *Resilient.* Women's preferences for these three concepts were only surpassed by *Intercropping*. Women in Uganda preferred *Home use* roughly equally to *Intercropping.* Both *Drought escape* and *Resilient* were more preferred by women in Uganda over *Home Use* and *Intercropping*. Among male farmers, the *Home use* concept scored significantly lower than the top concepts in the respective countries. Interestingly, the overall ranking of *Home use* in Kenya and Uganda was fairly similar while in Kenya approximately half of maize harvest is consumed while this is less than 5% in Uganda.

*Family nutrition* stood out in the terms of associated with it which was clearly differentiated from the other concepts. The positioning of Pro-vitamin A maize as a family nutrition did speak to a significant group of the consumers. *Food and fodder* also attracted a specific audience for whom livestock was probably more important.

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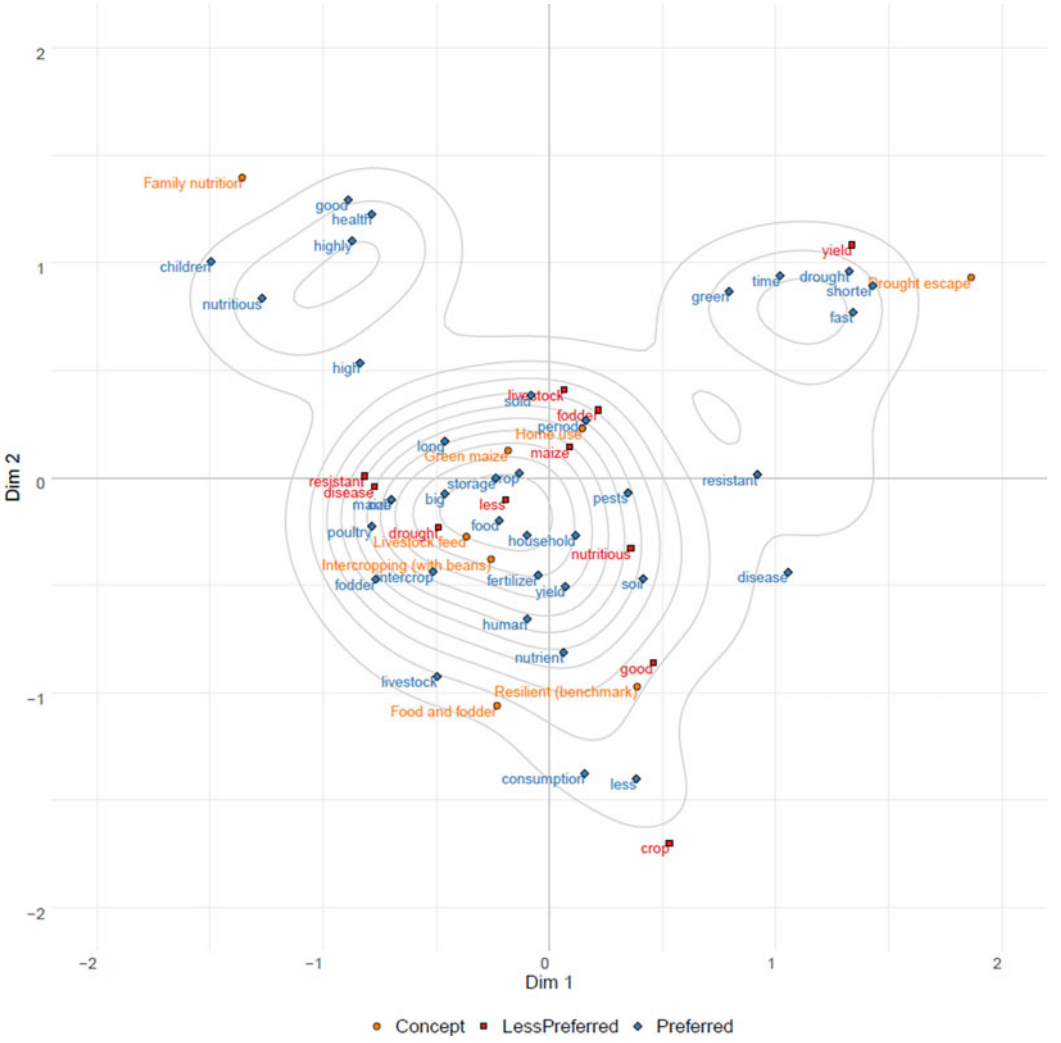



**Figure 2.** Correspondence of concepts with participants reasoning for preferring or not a given concept. *The distance between points represents relationships between words. The orange dots represent the maize product concepts. The blue diamonds represent words that were positively associated with the concept, meaning that meaning that they might be used to express an advantage of using the concept. The red squares represent the words negatively associated to the concept, meaning that they might be used to express a concern or disadvantage of using the concept.*

*Least preferred: Green maize and Livestock feed*

*Green maize* and lastly *Livestock feed* were not prioritized among the participants. Especially the *Feed* concept scored significantly lower than all other concepts in both countries across men and women. Remarkably, the yellow colour of the grain was not mentioned as the reason for its low performance, the key reason seemed to be the restricted purpose of livestock feed. *Green maize* also did not score well in Kenya, maybe because most maize in the target region is turned into maize flour for the popular local dish 'Ugali'. *Green maize* scored the same as *Family Nutrition* and *Food and Fodder* in Uganda and thus shower greater potential as a concept than in Kenya.

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**Table 5.** Head-to-head comparison (in probability to win) of pairwise comparisons of product concepts. Values in the left indicate the probability that one concept has in winning against the concept in the heading (e.g. drought avoidance has a 51% of winning against resilient)

| | Resilient (benchmark) | Drought escape | Food and fodder | Home use | Green maize | Livestock feed | Intercropping (with beans) | Family nutrition |
|---|---|---|---|---|---|---|---|---|
| Resilient (benchmark) | | 49% | 64% | 57% | 75% | 87% | 51% | 65% |
| Drought escape | 51% | | 65% | 58% | 76% | 88% | 52% | 66% |
| Food and fodder | 36% | 35% | | 43% | 63% | 80% | 37% | 52% |
| Home use | 43% | 42% | 57% | | 69% | 84% | 44% | 58% |
| Green maize | 25% | 24% | 37% | 31% | | 70% | 26% | 39% |
| Livestock feed | 13% | 12% | 20% | 16% | 30% | | 13% | 21% |
| Intercropping (with beans) | 49% | 48% | 63% | 56% | 74% | 87% | | 64% |
| Family nutrition | 35% | 34% | 48% | 42% | 61% | 79% | 36% | |

## Discussion

Our results identified three product concepts that clearly outperformed the others in terms of farmer preferences rankings: *Resilience*, *Drought escape*, and *Intercropping*. The strong showing for the *Resilience* concept confirmed the current breeding priority by CGIAR and NARES for East Africa focused on incremental yield gains under adverse smallholder growing conditions. The strong showings for the *Drought escape* and *Intercropping* concepts suggest that new opportunities exist for innovation in product design. Both potential future products are discussed in turn below.

### *Farmers in mid-altitude growing regions require drought escape products*

Public sector breeding programs for hybrid maize in East Africa have, for decades, focused on the largest share of their investments on intermediate maturity and drought tolerance hybrids for mid-altitude growing regions. Results from this study suggest that mid-altitude farmers seek out different types of seed products to deal with drought. For many of the farmers who participated in this study, the potential reduction in yield from the use of drought escape products would be compensated for by the flexibility to plant later in the season and reduced risks from drought stresses late in the season. Public breeding programs currently design and test early maturity hybrids only for dryland areas. These results suggest that public programs, as well as privately owned seed companies, should test early maturity hybrids in zones with higher rainfall and make these available to farmers.

Farmer interest in drought escape (early maturity) maize seed products has been confirmed by other studies in East Africa. Kogo *et al.* (2022) found that selecting early maturity seed products was one of the most popular adaptation strategies to climate change, together with changing planting dates and crop diversification for farmers in Western Kenya. An analysis of panel data of maize-seed sales from agrodealers in Kenya during 2020–2022 showed that during the long-rains season, farmers in higher rainfall production environments (wet, mid, and high altitudes) purchased early maturity seed products despite potentially lower yields from these products (Rutsaert *et al.* 2023). Interestingly, the *Drought escape* concept clearly outperformed all other concepts in Uganda.

### *Farmers prioritized products that supported intercropping practices*

A second key finding from this study was the strong farmer interest in the *Intercropping* concept. Over 80% of our sample in Kenya intercropped maize, mainly with beans. Seed companies in Kenya typically do not use language or imagery on intercropping to market their seed products.

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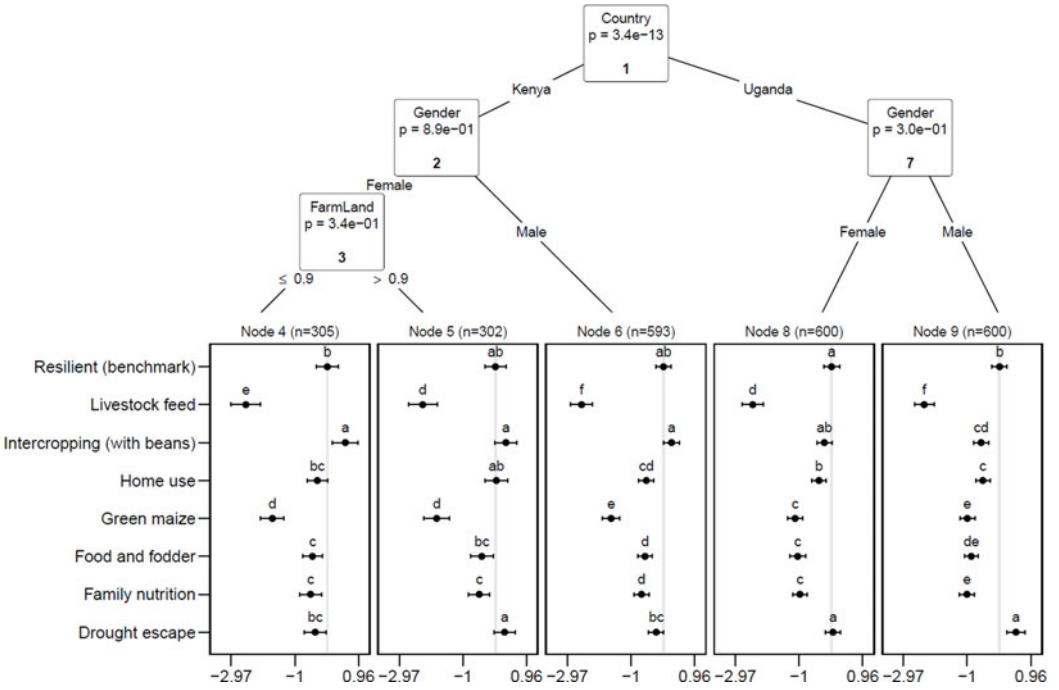



**Figure 3.** Model estimates (log-worth) of farmers' preferences on the product concepts of hybrid maize segmented by country, gender, and farmland using recursive partitioning trees. *The horizontal axis of each panel shows coefficients derived from the Plackett–Luce model, which estimates the probability of one given concept in outperforming all the other concepts in the set. Error bars show quasi-SEs. The grey vertical lines indicate the reference concept 'Resilient (benchmark)' (log-worth set to zero). Different letters indicate significant differences at* $p < 0.05$.

Furthermore, public maize breeding programs in Sub-Saharan Africa typically test the performance of pre-released varieties under monocropping conditions both on-station and on-farm (Davis & Woolley 1993; Huang et al. 2019; Pixley *et al.* 2023). Based on the results of this study, an obvious recommendation is that the testing of pre-commercialized hybrids be carried out under commonly used smallholder production practices, with intercropping primary among them.

Taking this finding a step further, breeding programs could actively select traits for 'superior intercrop performance' (Snapp & Silim, 2002). As suggested decades ago by Davis and Woolley (1993), the focus of crop breeding for an intercropping specific product should be on traits 'which enhance the complementary effect between species'. The interactions between plant genetics and smallholder cropping systems have been a longstanding field of research, recently summarized by Demie *et al.* (2022), which showed (i) that performance of intercropping systems depend on the genotypes used in mixture and (ii) that performance under intercropping was poorly correlated with performance under monocropping. These findings call for deeper discussions between social scientists, agronomists, and crop breeders on whether the current ideotype or plant architecture of currently available crops can be improved towards better suitability for different cropping systems. Second, these findings can help to rethink current protocols for testing new lines or crosses under station and farmer conditions. Third, the identification of unique essential traits for a potential intercropping seed product will require a better understanding of farmer expectations for plants grown under intercropping conditions.

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## Conclusion

This paper explored how to identify the potential for future innovation in the design of maize seed products in two East African countries where maize plays an important role in food security. To date, discussions among breeders and economists have tended to zoom in on farmers' current requirements for specific traits, mainly yield growth and stability. This paper took a different approach by seeking to understand the opportunities for future innovation in seed product design based on how farmers grow and use maize in the context of their livelihood pursuits. Our application of product concept testing in Uganda and Kenya for hybrid maize identified two clear opportunities for future investments in seed product design: drought escape in Uganda and intercropping in Kenya. Testing also identified several concepts with potential for short-term term impact if available and if product development were accompanied by investments in seed systems (e.g. retailer engagement, product marketing, and public sector purchasing): nutritious, home use, and food and fodder. Future engagements by breeders, social sciences, food technologists, and agronomists are needed to identify the required traits (and their thresholds) for these potential products.

**Supplementary material.** The supplementary material for this article can be found at https://doi.org/10.1017/S001447972500002X

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